2. Regression

Regression is used to detect or create a link between one or more dependent and independent variables.

2.1 Development of model

We are going to calculate the regression model with Excel data analysis.

- Y range will be shelf life in days that is Dependent Variable (DV)
- X range will be Mean of NH4, temperature & humidity that is Independent Variable (IV).
- Confidence Level is 99% (Alpha = 0.01)

Regression Statistics	
Multiple R	0.98
R Square	0.95
Adjusted R Square	0.95
Standard Error	0.56
Observations	81.00

If P > 0.01(Significant Value) that product will affect your shef-life

Only Blackspot impacts Shelf-life (Days)

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*	
Regression	4.00	493.35	123.34	387.19	0.00	Lesser than P Value
Residual	76.00	24.21	0.32			
Total	80.00	517.56				

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 99.0%*	*Upper 99.0%*
Intercept	15.88	0.26	61.99	0.00	15.37	16.39	15.21	16.56
MKRH	(0.15)	0.01	(22.93)	0.00	(0.17)	(0.14)	(0.17)	(0.14)
MKT	(0.06)	0.01	(10.46)	0.00	(0.07)	(0.05)	(0.07)	(0.04)
NH4	(0.07)	0.01	(11.52)	0.00	(0.08)	(0.05)	(0.08)	(0.05)
BlackSpot	(0.08)	0.22	(0.36)	0.72	(0.53)	0.36	(0.67)	0.51

Table 5. Regression Excel Output

2.2 Interpretation of Regression Results

- P-Value > Significant level (Alpha) on our Dependent Variable, it is outliner in this model, so we are not considering it as Independent Variable in our analysis.
- P-Value < Significant Value (Alpha), it has significant in our Dependent Variable.
- R-square (0.95) is strength of linear regression model developed, it signifies high dependability of shelf days on NH4, temperature & humidity.
- Sum of square (493.35) signifies the total variation of Dependent Variable regression model has attained to run.
- Significance F signifies the dependability of variables, less the value higher the significance.

- Coefficients signifies relation level between IV and DV that is MKRH (-0.15), MKT (-0.06) & NH4 (-0.07) has impact on shelf life, temperature & NH4. Here negative indicates that if IV increases then DV decreases that is days (shelf life).

2.3 Normality Distribution

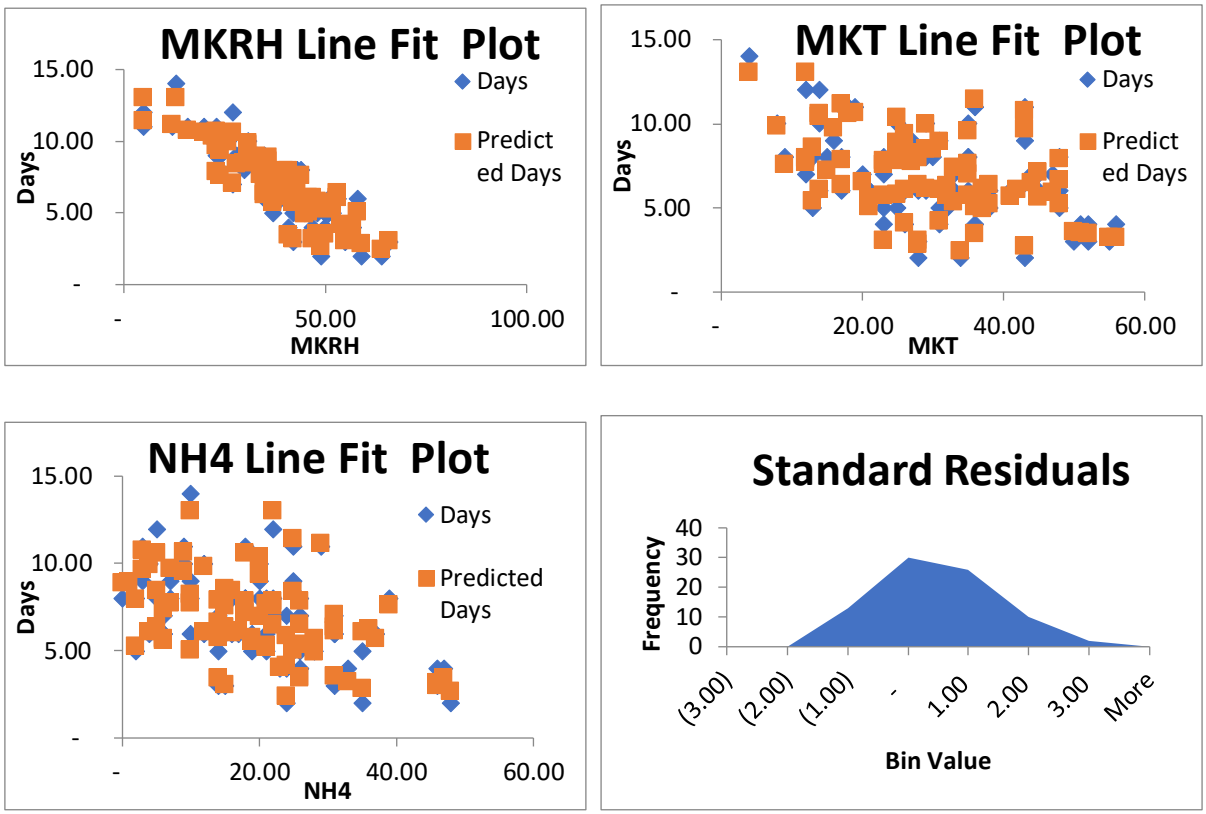

Fig 3. Charts of Regression Results

For any value of x, we will have multiple value of y and will be normally distributed because normality assumptions in random errors are statistically independent. Errors will have equal variances for all IV. Population regression model will pass through DV for all values of IV. Coefficient available in our regression model the IV has inverse effect on DV that is if we decrease our IV, then DV increases and vice versa. So, if we decrease our MKRH, MKT, NH4, all the three variables have negative coefficient so it will have inverse effect on shelf-life days. From standard residuals chart we can clearly understand that residuals are distributed in standard way, so our normality assumptions are correct.

2.4 Model Improvement

Organisation can reduce the mean kinetic temperature & humidity and should try to minimize the NH4 content within the product to increase the level of shelf days and for better consumption of resources.